FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – III
PAPER –2
TEST DATE: 25-12-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. B

Sol. If $A > 2\theta_C$ then light does not emerge.

$A = 60^o$

θ_C should be less than 30°

$$\sin 30° = \frac{1}{\mu}$$

$$\Rightarrow \quad \mu = 2$$

minimum value of μ is 2.

2. B

Sol. $$\lambda = \frac{h}{mv}$$

$$\frac{1}{2}m\bar{v}^2 = \frac{3}{2}k_BT$$

$$v_{rms} = \sqrt{\bar{v}^2} = \sqrt{\frac{3k_BT}{m}}$$

$$\lambda = \frac{h}{mv_{rms}} = \frac{h}{\sqrt{3mk_BT}}$$

3. A

Sol. From Moseley's law,

$$\frac{1}{\lambda_{K_\alpha}} \propto (Z-1)^2$$

$$\frac{\lambda_{M_0}}{\lambda_{Cu}} = \frac{(Z_{Cu}-1)^2}{(Z_{M_0}-1)^2} = \frac{(28)^2}{(41)^2}$$

$$\lambda_{M_0} = \lambda_{Cu} \times \frac{(28)^2}{(41)^2} = 0.71Å.$$

4. B

Sol. $N_{238}\lambda_{238} = N_{234}\lambda_{234}$

$$\frac{N_{238}}{T_{1/2}(^{238}U)} = \frac{N_{234}}{T_{1/2}(^{234}U)}$$

$$T_{1/2}(^{234}U) = \frac{N_{234}}{N_{238}} \times T_{1/2}(^{238}U)$$

$$= \frac{1}{1.8\times10^4} \times 4.5\times10^9 \text{ y} = 2.5\times10^5 \text{ yr}$$

5. B, C, D

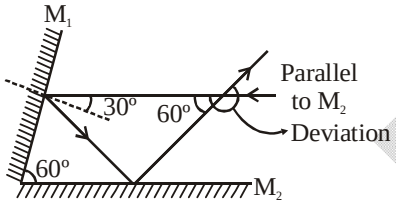
Sol. For the critical ray at the interface,

$\mu_2 \sin i = \mu_1 \sin 90^o$

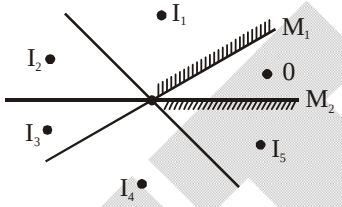
$$\theta_1 = (90^o - i) \equiv \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}.$$

6. A, B, C, D

Sol. (A)

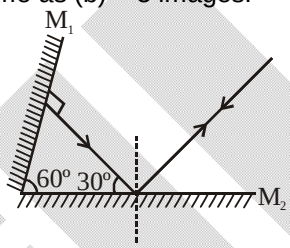


(B)



(C) same as (b) – 5 images.

(D)



7. A, D

Sol. For equilibrium, the system must be in a local minima of potential energy. This is possible only if the centre of mass of the system lies below O i.e. the centre of the hemisphere. In the critical case, the centre of mass of the system can coincide with O.

Now taking O as the origin, say H is the maximum principle height of the cone then:

$$\left(-\frac{3R}{8}\right)\cdot\frac{2}{3}\pi R^3 + \left(\frac{H}{4}\right)\cdot\frac{1}{3}\pi R^2 H = 0$$

$$\Rightarrow \quad -\frac{R^2}{4} + \frac{H^2}{12} = 0 \quad \text{or} \quad H = \sqrt{3}R$$

8. B, C

Sol. Initial pressure at the bottom = $\rho g \times 2H + 2\rho \times g \times H = 4\rho gH$.

Final density of the homogenous mixture:

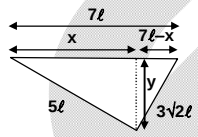
$$= \frac{\rho \times A \times 2H + 2\rho \times 2A \times H}{A \times 2H + 2A \times H} = \frac{3}{2}\rho$$

Final pressure $= \frac{3}{2}\rho \times g \times 3H = \frac{9}{2}\rho gH$.

9. B, C

Sol. $y^2 = (5\ell)^2 - x^2 = (3\sqrt{2}\ell)^2 - (7\ell - x)^2$

$\Rightarrow \quad x = 4\ \ell \Rightarrow y = 3\ell \Rightarrow T = 2\pi\sqrt{\frac{3\ell}{g}}$



10. B, D

Sol. $\frac{1}{2}mv^2 + \left(-\frac{2GMm}{L}\right) = 0$

$\Rightarrow v = 2\sqrt{\frac{GM}{L}}$

Gravitational field is conservative, in which total mechanical energy (KE + PE) remains conserved. Hence, the kinetic energy imparted to the mass m is gradually reduced and gets converted into its potential energy, so that at every point of its fl ight the total mechanical energy remains constant.

Section – B

11. 1

Sol. $H_{app} = \frac{H}{\mu}\left(\frac{\cos i}{\cos r}\right)^3 = \frac{9}{\sqrt{3}} \times \left(\frac{1}{\sqrt{3}}\right)^3 = 1\text{m}$

12. 2

Sol. Let be λ is wavelength of sound in air

$\ell_1 = \frac{\lambda}{4}$ and $\ell_2 = \frac{3\lambda}{4}$

$\lambda = 2(\ell_2 - \ell_1)$

$v = f\lambda = 165 \times 2(150 - 50) \times 10^{-2} = 330$ m/s

$\frac{dv}{v} = \frac{|d\ell_2| + |d\ell_1|}{\ell_2 - \ell_1}$

$dv = \frac{0.5 + 0.1}{150 - 50} \times 330 = 1.98$ m/s

$v = (330 \pm 2)$ m/s, it is speed of sound in air.

13. 5

Sol. Let equation of the wave be

$y = A \sin kx \cos\omega t$

$\because \quad 5\sqrt{3} = A \sin(k \times 10) \qquad \ldots (1)$

and $5\sqrt{3}$ = A sin(k×20) . . . (2)

From (1) and (2), and the given conditions,

$20k = \pi - 10k \Rightarrow k = \frac{\pi}{30}$

$5\sqrt{3} = A \sin\left(\frac{\pi}{30}\times 10\right)$ using equation (1)

$\Rightarrow$ A = 10mm

$\because k = \frac{2\pi}{\lambda} = \frac{\pi}{30}$,

$\Rightarrow \lambda$ = 60cm.

$\because L = n\frac{\lambda}{2}$

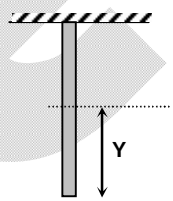
$\Rightarrow n = \frac{2L}{\lambda} = \frac{2\times 150}{60} = 5$

As the number of loops is 5, the string vibrates in the 4th overtone 5th harmonic.

14. 4

Sol. Tension in the string at a height y from the lower end

= Weight of the lower part (having y length) = μyg



Wave speed at this point V= $\sqrt{\frac{T}{\mu}}$

$\Rightarrow V = \sqrt{\frac{\mu yg}{\mu}} = \sqrt{yg}$

$\Rightarrow \frac{dy}{dt} = \sqrt{yg}$

On integration we get,

$y = \frac{gt^2}{4}$

- Distance travelled by the body in time $t = \frac{1}{2}gt^2$

  When the pulse & the body are at same level

  $\frac{gt^2}{4} + \frac{1}{2}gt^2 = L$

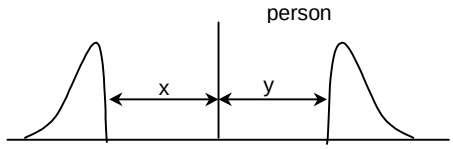
  $\Rightarrow t = \sqrt{\frac{4L}{3g}}$

Therefore, distance from the top

$= \frac{1}{2}gt^2 = \frac{2L}{3}$

15. 4

Sol. $t_1 = \frac{2x}{v}$ & $t_2 = \frac{2y}{v}$



$\therefore x + y = \frac{v}{2}(t_1 + t_2) = \frac{332}{2}[1.5 + 2.5] = 664$ m.

third echo will be heard after

$t_3 = t_1 + t_2 = 1.5 + 2.5 = 4$ sec.

16. 5

Sol. $\frac{P_0^2}{2\rho V} = I = \frac{\text{Power}}{4\pi r^2}$

Put the values to get the answer.

17. 4

Sol. $(\pi r^2)\sqrt{2gy} = \pi x^2\left(-\frac{dy}{dt}\right)$

$\Rightarrow (r^2)\sqrt{2gy} = x^2\lambda$

$\Rightarrow y \alpha x^4$

18. 4

Sol. $Y = \frac{FL}{A\ell}$

Here measurement is for ℓ only,

So, $\frac{\Delta Y}{Y} = \frac{\Delta \ell}{\ell}$

From observation, ℓ_1 = MS + 20(LC), and ℓ_2 = MS + 45(LC)

Change in length = $\ell_2 - \ell_1$ = 25 $\times$ LC, and the maximum permissible error in measurement of elongation in one LC.

$\frac{\Delta Y}{Y} \times 100\% = \frac{1}{25}\frac{LC}{(LC)} \times 100\% = 4\%$

Chemistry

PART – II

Section – A

19. D

Sol. $IF_3 \rightarrow$ Yellow powder

$IBr \rightarrow$ Black solid

$ICl \rightarrow$ Red solid

$IF_7 \rightarrow$ Colourless gas

20. C

Sol. $\Delta S_{mix} = -n_{total} R \sum x_i \ell n x_i$

$n_{CO_2} = 1 \qquad n_{N_2} = 1 \qquad n_{Ar} = 2$

$\therefore \chi_{CO_2} = \frac{1}{4} \qquad \chi_{N_2} = \frac{1}{4} \qquad \chi_{Ar} = \frac{1}{2}$

$$\Delta S_{mix} = -4R\left\{\frac{1}{4}\ell n\frac{1}{4} + \frac{1}{4}\ell n\frac{1}{4} + \frac{1}{2}\ell n\frac{1}{2}\right\}$$

$$= -4R\left\{\frac{1}{2}\ell n\frac{1}{4} + \frac{1}{2}\ell n\frac{1}{2}\right\}$$

$$= -4R \times \frac{3}{2}\ell n\frac{1}{2}$$

$$= -6R\ell n\frac{1}{2} = 6R\ell n2$$

21. D

Sol. $\Lambda_m = \Lambda_m^o - b\sqrt{C}$ … (i)

When it is diluted $C' = \frac{C}{16}$

$\Lambda'_m = \Lambda_m^o - b\sqrt{\frac{C}{16}}$ … (ii)

$$\therefore \left|\Lambda_m - \Lambda'_m\right| = \left|-b\sqrt{C} + \frac{b\sqrt{C}}{4}\right|$$

$$= \frac{3b\sqrt{C}}{4}$$

22. D

Sol. ${}_{64}Gd \rightarrow [Xe]4f^7 5d^1 6s^2$

${}_{64}Gd^{+2} \rightarrow [Xe]4f^7 5d^1 6s^0$

23. B, C

Sol. 1 equivalent of $H_2SO_4 = 1$ equivalent of $H-$atom

$V_{H_2O_2} = 5.6 \times \text{Normality} = 5.6 \times \frac{1}{5} = 1.12$

$C_2H_4 + 3O_2 \longrightarrow 2CO_2 + 2H_2O$

$\therefore$ Moles of O_2 required $= \frac{14}{28} \times 3 = \frac{3}{2}$

$\therefore$ Weight of $O_2 = \frac{3}{2} \times 32 = 48$ g

Moles of 'O' atom in 276 gm of $N_2O_4 = \frac{276}{92} \times 4 = 12$ mole.

24. A, B, D

Sol. For isotonic solution, osmotic pressure (π) should be same.

$\pi = i.CRT$

i.C for 0.2 M NaCl $(\alpha = 1)$

$= \left[1 + (x + y - 1)\alpha\right] C$

$= \left[1 + (1 + 1 - 1)1\right] 0.2 = 0.4$

i.C for 0.2 M $MgCl_2$ $(\alpha = 0.5)$

$= \{1 + (1 + 2 - 1) \times 0.5\} \times 0.2$

= 0.4

i.C for 18 g glucose in 250 ml solution $= 1 \times \frac{18/180}{250/1000} = 0.4$

i.C for 0.5 mole of $Al(NO_3)_3$ in 4 litre $(\alpha = 0.4)$ $= \{1 + (1 + 3 - 1)0.4\} \frac{0.5}{4}$ = 0.275

i.C for 1 mole of $AlCl_3$ $(\alpha = 0.2)$ in 4 litre solution $= \{1 + (1 + 3 - 1)0.2\} \times \frac{1}{4} = 0.4$

$\therefore$ A, B and D are isotonic solution with 0.2 M NaCl $(\alpha = 1)$.

25. A, B, D

Sol. Strength of M – C back bonding

α electronic density of metal

α M – C bond order

α C – O bond length

M – C back bonding order

$\left[Ti(CO)_6\right]^{2-} > \left[V(CO)_6\right]^{-} > \left[Mn(CO)_6\right]^{+} > BH_3.CO$

$\therefore$ Bond length order z > y > x > w.

26. A, C

Sol. $\left(\frac{dP}{dT}\right)_V = \frac{d}{dT}\left(\frac{RT}{V}\right)_V = \frac{R}{V}$

$\left(\frac{dV}{dT}\right)_P = \frac{d}{dT}\left(\frac{RT}{P}\right)_P = \frac{R}{P}$

$\therefore \left(\frac{dP}{dT}\right)_V \times \left(\frac{dV}{dT}\right)_P = \frac{R}{V} \times \frac{R}{P} = \frac{R^2}{PV} = \frac{R^2}{RT} = \frac{R}{T}$

27. A, B

Sol. PH_3 – Toxic

$P_4 + 3NaOH + 3H_2O \longrightarrow PH_3 + 3NaH_2PO_2$

NH_3 is more basic than PH_3

N – H bond energy (389 kJ/mol) > P – H bond energy (318 kJ/mol)

28. B, C

Sol. Mg^{2+} belongs to Group-VI.

$Hg^{+2} + KI \longrightarrow K_2[HgI_4]$

Option (A) and (D) are wrong.

Section – B

29. 7

Sol. Group – V cations are $Ca^{+2}, Sr^{+2}, Ba^{+2}$.

30. 5

Sol. White $\rightarrow Ag_2CO_3$, Ag_2SO_3, $AgNO_2$, ZnS, $BaCO_3$

Black $\rightarrow Ag_2S$, FeS, PbS

Yellow $\rightarrow$ AgI

Reddish brown $\rightarrow Fe(OH)_3$

31. 5

Sol. Fe_3O_4 is an example of inverse spinel, in which one Fe^{+3} ion present at tetrahedral void and one Fe^{+3} as well as one Fe^{+2} ion present at octahedral voids per unit cell.

$\therefore$ 1 Fe^{+3} ion is present at tetrahedral voids.

$Fe^{+3}\left(d^5\right) \rightarrow e^2t_2^3$

$\therefore$ Total 5 unpaired electrons.

32. 6

Sol. Cerrusite $\rightarrow PbCO_3$

Azurite $\rightarrow Cu(OH)_2.2CuCO_3$

Calamine $\rightarrow ZnCO_3$

Zincite $\rightarrow$ ZnO

Siderite $\rightarrow FeCO_3$

Magnetite $\rightarrow Fe_3O_4$

Magnesite $\rightarrow MgCO_3$

Dolomite $\rightarrow MgCO_3$, $CaCO_3$

Bauxite $\rightarrow Al_2O_3.2H_2O$

33. 4

Sol. $Fe_4\left[Fe(CN)_6\right]_3$, $K_2Cr_2O_7$, MnO_4^-, V_2O_5 are coloured due to charge transfer.

34. 5

Sol. 1. Pressure and density are intensive property as it does not depends on amount.
2. Every close system is not isolated system.
3. $\Delta S = 0$ for adiabatic reversible process.
4. ΔG should be – ve for a spontaneous processs.
5. For cyclic process $\Delta U = 0$.
6. $\frac{V_2}{V_1} = \frac{T_2}{T_1} = \frac{573}{473}$.

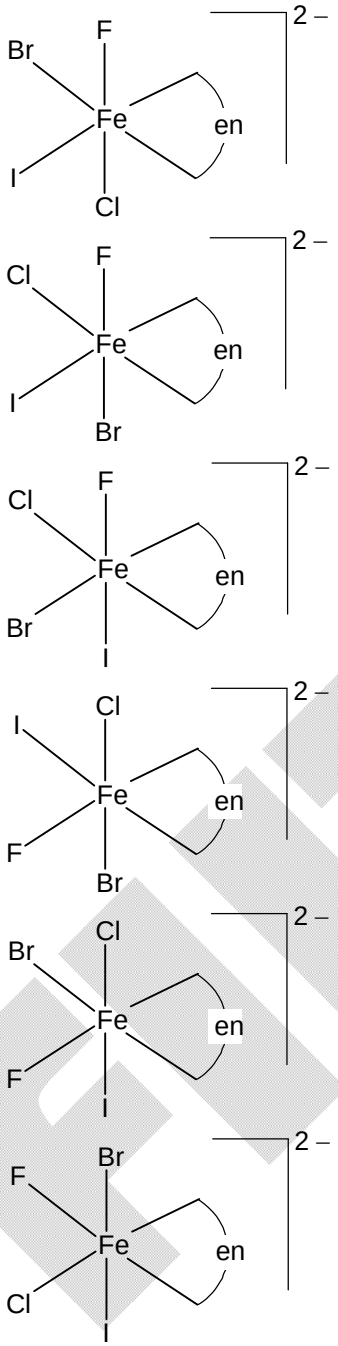
7. Calorific value of $C_2H_6 = \frac{341.1}{30} = 11.37$ kcal / g . Calorific value of $C_2H_2 = \frac{310}{26} = 11.92$ kcal / g .

$\therefore$ Ethyne is a better fuel.

1, 3, 4, 5 and 7 are correct statement.

35. 6

Sol.



36. 3

Sol. Ertythromycin, Tetracyclin and Chloramphenical has inhibitory effect on microbes.

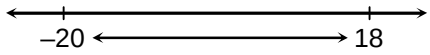
Mathematics

PART – III

Section – A

37. D

Sol. By rational root theorem turn the given polynomial into a degree = 2 polynomial with rational root



x = 1, –1, 2, –2 are all possible root

So, x = –2, x = 1. Polynomial is equal to 0

So, polynomial becomes $(x - 1)(x + 2)(x^2 + (2a - 1)x + 1)$

So, we only need to focus on the quadratic

$(2a - 1)^2 - 4 \geq 0 \Rightarrow n(S) = 38$; $n(E) = 36$ $\therefore$ Probability = $\frac{36}{38} = \frac{18}{19}$

38. B

Sol. Let $x_3 = 3$, then $x_1x_2x_3 + x_2x_3x_4 + x_3x_4x_5 + x_4x_5x_1 + x_5x_1x_2 = 3(x_1x_2 + x_2x_4 + x_4x_5) + x_5x_1(x_2 + x_4)$

So, the sum is divisible by 3 if $x_2 + x_4$ is divisible by 3, the possible sums of $(x_2 + x_4)$ can be (1, 2), (2, 1), (1, 5), (5, 1), (2, 4), (4, 2), (4, 5) or (5, 4) are we have 2 ways to designate x_1 and x_5 for a total of $8 \times 2 = 16$

So, desired = $16 \times 5 = 80$

39. D

Sol. $z^3 - 8 = 0 \Rightarrow (z - 2)(z^2 + 2z + 4) = 0 \Rightarrow z = 2$ and $z^2 + 2z + 4 = 0 \Rightarrow z = -1 \pm \sqrt{3}\,i$

and $z^3 - 8z^2 - 8z + 64 = 0 \Rightarrow (z^2 - 8)(z - 8) = 0 \Rightarrow z = \left(2\sqrt{2}, -2\sqrt{2}, 8\right)$

Maximum distance is from 8 and $-1 \pm \sqrt{3}\,i$ i.e. $2\sqrt{21}$

40. B

Sol. $[x]\{x\} = ax^2$

Case-I: $[x] \cdot \{x\} < 0$ where $x < 0$ while $ax^2 \geq 0\ \forall\ x \in R$ and $x = 0$ is one solution but that does not affect the sum of the solution

Thus we need to look at (+ve) solutions for $n \leq x < n + 1$

$[x] = n$, $\{x\} = x - n$ equation is $nx - n^2 = ax^2$

$ax^2 - nx + n^2 = 0 \Rightarrow x = \frac{n \pm \sqrt{n^2 - 4an^2}}{2a} = n\left(\frac{1 \pm \sqrt{1 - 4a}}{2a}\right)$

For (+ve) roots we consider $x = \frac{n}{2a}\left(1 - \sqrt{1 - 4a}\right)$

Since, sum of all solutions is 420

$\Rightarrow \sum \frac{n}{2a}\left(1 - \sqrt{1 - 4a}\right) = 420 \Rightarrow \left(\frac{1 - \sqrt{1 - 4a}}{2a}\right) \cdot \sum n = 420$

$\Rightarrow \left(\frac{1 - \sqrt{1 - 4a}}{2a}\right)\left(\frac{n(n+1)}{2}\right) = 420 \Rightarrow \left(\frac{1 - \sqrt{1 - 4a}}{2a}\right) \cdot 406 = 420$

(Considering n is 28 i.e. $\frac{n(n+1)}{2} = 406$ at n = 28 which is nearer to 420)

$\Rightarrow a = \frac{29}{900}$

41. A, B, C

Sol. Let $Q(x) = P(x^2 + 4x - 7) \Rightarrow Q(1) = P(-2) = 0 \Rightarrow P(x) = a(x + 2)(x - \alpha)$

Now, $Q(x) = P(x^2 + 4x - 7) = a(x^2 + 4x - 5)(x^2 + 4x - 7 - \alpha)$

$\Rightarrow Q(x) = a(x-1)(x+5)(x^2+4x-(7+\alpha))$
As Q(x) has at least one repeated root, this must be x = 1 or x = –5
$Q(x) = a((x-1)(x+5))^2$ or $x^2+4x-(7+\alpha)=0$ has double roots if D = 0, $\alpha = -11$
$Q(x) = a(x-1)(x+5)(x+2)^2$

42. A, C

Sol. $b_n = \sum_{r=0}^{n}\frac{1}{{}^nC_r} = \sum_{r=0}^{n-1}\frac{1}{{}^nC_r}+\frac{1}{{}^nC_n} = \sum_{r=0}^{n-1}\frac{(n-r)!r!}{n!}+1 = \sum_{r=0}^{n-1}\left(\frac{(n-r)}{n}\right)\left(\frac{((n-1)-r)!r!}{(n-1)!}\right)+1$

$\Rightarrow \frac{1}{n}\cdot\sum_{r=0}^{n-1}\frac{(n-r)}{{}^{n-1}C_r}+1$ ..... (1)

$\Rightarrow \frac{1}{n}\cdot\sum_{r=0}^{n-1}\frac{n-(n-1-r)}{{}^{n-1}C_{n-1-r}}+1$

$\Rightarrow \frac{1}{n}\cdot\sum_{r=0}^{n-1}\frac{r+1}{{}^{n-1}C_r}+1$ ..... (2)

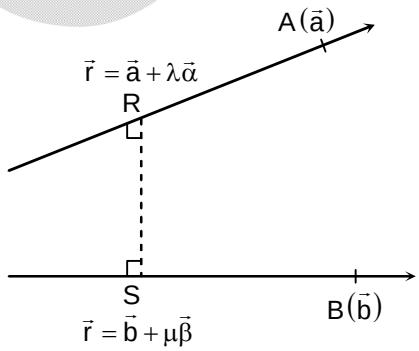
Adding equation (1) and (2), we get $2b_n = \frac{1}{n}\cdot\sum_{r=0}^{n-1}\frac{n-r+r+1}{{}^{n-1}C_r}+2$

$\Rightarrow \frac{n+1}{2n}\cdot\sum_{r=0}^{n-1}\frac{1}{{}^{n-1}C_r}+1 \Rightarrow \frac{n+1}{2n}\cdot b_{n-1}+1$

43. B, D

Sol. Let position vector R is $\vec{a}+\lambda\vec{\alpha}$ and position vector of S is $\vec{b}+\mu\vec{\beta}$

Vector $\overrightarrow{RS} = (\vec{a}-\vec{b})+\lambda\vec{\alpha}-\mu\vec{\beta}$ taking dot product of $\overrightarrow{RS}$ with $\vec{\alpha}$ and $\vec{\beta}$ we will get λ or μ and then the position vector R and S



44. B, D

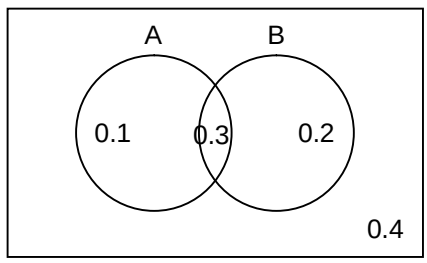
Sol. P(A) = 0.4, $P(A\cup B) = P(A)+P(B)-P(A\cap B) = 0.6$
P(B) = 0.5, $P(A\cap B) = 0.3$

$P(E_1) = \frac{P(A\cap B)}{P(A\cup B)} = \frac{0.3}{0.6} = \frac{1}{2}$

$P(E_2) = \frac{P(A\cap\bar{B})+P(\bar{A}\cap B)}{P(A\cup B)} = \frac{0.1+0.2}{0.6} = \frac{1}{2}$

$P\left(\frac{A}{B}\right) = \frac{P(A\cap B)}{P(B)} = \frac{0.3}{0.5} = \frac{3}{5}$; $P\left(\frac{B}{A}\right) = \frac{0.3}{0.4} = \frac{3}{4}$

$P\left(\frac{A}{A\cup B}\right) = \frac{P(A)}{P(A\cup B)} = \frac{0.4}{0.6} = \frac{2}{3}$; $\frac{P(B)}{P(A\cup B)} = \frac{0.5}{0.6} = \frac{5}{6}$



45. B, D

Sol. A_n is non-singular for all n and $\det A_n = (-1)^{n+1}\cdot n!$

46. A, B, C, D

Sol. For $r \geq 0$, $\frac{1}{r+1} \cdot C_2^2 = \frac{1}{(r+1)} \cdot \frac{n!}{r!(n-r)!} \cdot C_r = \frac{1}{(n+1)} \cdot \frac{((n+1)!)}{(r+1)!(n-r)!} \cdot C_r = \frac{1}{(n+1)} \cdot {}^{(n+1)}C_{r+1} \cdot {}^{n}C_r$

Thus, $S = \frac{1}{(n+1)} \cdot S_1$ where $S_1 = \sum_{r=0}^{n} {}^{n+1}C_{r+1} \cdot {}^{n}C_r = \frac{(2n+1)!}{n!(n+1)!}$

$\therefore\ S = \frac{1}{n+1} \cdot \frac{(2n+1)!}{n!(n+1)!} = \frac{(2n+1)!}{((n+1)!)^2} = \frac{1}{(2n+2)} \cdot {}^{2n+2}C_{n+1}$

Section – B

47. 2

Sol. Let $x = \frac{y}{\sqrt{2014}}$ i.e. is $\frac{y^3}{2014} - \frac{4029y^2}{2014} + 2 = 0 \Rightarrow y^3 - 4029y^2 + 4028 = 0$

$\Rightarrow (y - 1)(y^2 - 4028y - 4028) = 0$ it is observed as y = 1, is middle root

Therefore, $x_2(x_1 + x_3) = \frac{1}{\sqrt{2014}} \times \frac{4028}{\sqrt{2014}} = 2$

48. 7

Sol. Let $2^{17} = x$ given equation is $\frac{x^{17}+1}{x+1} = x^{16} - x^{15} + x^{14} + ... - x^1 + x^0$

Now, consider $(x^{16} - x^{15}) = x^{15}(x - 1) = x^{15}(2^{17} - 1)$

Since, $2^{17} - 1 = 2^{16} + 2^{15} + ... + 1 \Rightarrow$ Total 17 terms

So, there are 8 such pair like $x^{14} - x^{13}$ and so on

So, total is $17 \times 8 + 1 = 137$, 1 is for x^0

49. 1

Sol. As a subset with less than 3 chairs cannot contain 3 adjacent chair, these are only 10 sets of 3 chairs so that they are all 3 adjacent there are 10 subsets of 4 chairs where all r adjacent and $10 \times 5 = 50$, where there are only 3

If there are 5 chairs, 10 have all 5 adjacent $10 \times 4 = 40$ have 4 adjacent and $10 \times {}^5C_2 = 100$ have 3 adjacent with 6 chairs in the subset

10 have all 6 adjacent, $10 \times 3 = 30$ have 5 adjacent, $10 \times {}^4C_2 = 60$ have 4 adjacent

$\frac{10 \times 3}{2} = 15$ have 2 groups of 3 adjacent chairs and $10 \times \left({}^5C_2 - 3\right) = 70$ have 1 group of 3 adjacent chairs. All possible subsets with more than 6 chairs have at least 1 group of 3 adjacent. So adding all it is total = 581

50. 7

Sol. **First Solution:**

$(2+i)^n = \left(a_n + b_n\right)^i$ square the given equality $(3+4i)^n = \left(a_n + b_n i\right)^2 = \left(a_n^2 - b_n^2\right) + 2a_n b_n i$

So, $a_n b_n = \frac{1}{2} \text{Im}(3+4i)^n \Rightarrow \sum_{n=0}^{\infty} \frac{a_n b_n}{7^n} = \sum \frac{1}{2} \frac{\text{Im}(3+4i)^n}{7^n} = \frac{1}{2} \text{Im}\left(\frac{1}{1 - \frac{3+4i}{7}}\right) = \frac{7}{16}$

Second Solution:

$2 + i = \sqrt{5}\left(\frac{2}{\sqrt{2}} + \frac{1}{\sqrt{5}} i\right) = \sqrt{5}(\cos\theta + i\sin\theta) \Rightarrow \theta = \tan^{-1}\frac{1}{2}$

$(2+i)^n = \left(\sqrt{5}\right)^n (\cos\theta + i\sin n\theta) \Rightarrow \sum_{n=0}^{\infty} \frac{a_n b_n}{7^n} = \sum_{n=0}^{\infty} \frac{\cos n\theta \cdot \sin n\theta \cdot 5^n}{7^n}$

$= \frac{1}{2}\sum\left(\frac{5}{7}\right)^n \cdot \sin 2n\theta = \frac{1}{2}\text{Im}\sum\left(\frac{5}{7}\right)^n \cdot e^{i2n\theta} = \frac{1}{2}\text{Im}\left[\sum_{n=0}^{\infty}\left(\frac{5}{7}\right)^n (e^{i2\theta})^n\right]$ is a Geometric Progression

$\Rightarrow \frac{1}{2}\text{Im}\left[\frac{1}{1-\frac{5}{7}\cdot e^{2\theta i}}\right] \Rightarrow \frac{1}{2}\text{Im}\left(\frac{1}{1-\frac{5}{7}(\cos 2\theta + i\sin 2\theta)}\right) = \frac{1}{2}\text{Im}\left(\frac{7}{8}+\frac{7}{8}i\right) = \frac{1}{2}-\frac{7}{8} = \frac{7}{16}$

51. 9

Sol. The given sequence 4, 7, 1, 8, 9, 7, 6, 3, 9, 2, 1, 3, 4, 7, 1 .....
So, after 12th terms the separation of numbers starts and the sum of 12th terms is
4 + 7 + ... + 1 + 3 = 60

Since, $S_n > 10{,}000 \Rightarrow \frac{10{,}000}{60} = 166$ and $166 \times 12 = 1992$

So, $S_{1992} = 60 \times 166 = 9960 \Rightarrow 9960 + 4 + 7 + 1 + 8 + 9 + 7 = 9996 + 6 = 10{,}000$
$\therefore\ S_{1999} > 10{,}000$

52. 4

Sol. Let $\omega = \cos\alpha + i\sin\alpha$; $z = 10(\cos\beta + i\sin\beta)$

$\Rightarrow \frac{\omega - z}{z} = \frac{1}{10}\cos(\alpha-\beta) + \frac{1}{10}i\sin(\alpha-\beta) - 1$. Let $x = \alpha - \beta$

$\tan\theta = \frac{\sin x}{\cos x - 10}$, maximum occurs at $\cos x = \frac{1}{10}$, $\sin x = \pm\frac{\sqrt{99}}{10}$

$\therefore\ \tan^2\theta = \frac{1}{99}$

53. 3

Sol. Set of these rational numbers in from $\frac{1}{999}$ to $\frac{499}{501}$

There are 499 numbers between 501 and 999 $\Rightarrow$ 249 numbers are divisible by 2
$\Rightarrow$ 99 numbers are divisible by 5 $\Rightarrow$ 49 numbers are divisible by 10
So, by inclusion/exclusion 499 – 249 – 99 + 49 = 200

54. 7

Sol. $\ell = \lim_{n\to\infty} 18\left(\frac{3}{3^2}+\frac{3^2}{4^4}+\frac{3^3}{3^6}+.....\right) = 18\times\frac{1}{3\left(1-\frac{1}{3}\right)} = 9$

$m = \lim_{n\to\infty} 12\left(\frac{2}{2^2}+\frac{2^2}{2^4}+.....\right) = 12\times\frac{1}{2\left(1-\frac{1}{2}\right)} = 12 \Rightarrow A_n = \frac{9+12}{3} = 7$